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https://drive.google.com/file/d/1kZcXzONuM9N2ZCiCdi7Zd8BfPbEmXWCR/view?usp=drive_link

Automating Data Import and Relationship Mapping Using Import Sets & Dot Walking

Problem Statement:

Organizations frequently handle large volumes of external data originating from diverse sources such as CSV files, Excel spreadsheets, and databases. Manual entry of this data into platforms like ServiceNow is inefficient, time-consuming, and highly susceptible to human error.

Furthermore, maintaining accurate relationships between data entities—such as associating users with departments or reporting managers—becomes increasingly complex without a structured and automated approach. The absence of automation often results in data inconsistencies, duplication of records, and challenges in efficiently retrieving related information, ultimately impacting data integrity and operational efficiency.

Objective:

The main objective of this project is to automate the process of importing external data into ServiceNow and ensure accurate relationship mapping between records.

Key goals include:

- Automating data import using Data Sources
- Using Import Sets as a staging area for incoming data
- Creating Transform Maps to map and transfer data to target tables
- Establishing relationships between records (e.g., user → department, manager)
- Utilizing Dot Walking to easily access related data fields
- Improving data accuracy, consistency, and efficiency
- Reducing manual effort and errors in data handling

Skills & Technologies Used:

- ServiceNow Platform Fundamentals
- Data Import & Integration Techniques
- Working with Data Sources (CSV, Excel, Database)
- Import Sets & Transform Maps

- Field Mapping & Data Transformation Logic
- Reference Fields & Relationship Mapping
- Dot Walking for accessing related data
- Problem Solving & Automation Thinking

TASK INVITATION

Milestone 1: Import Data from Data Source

Description:

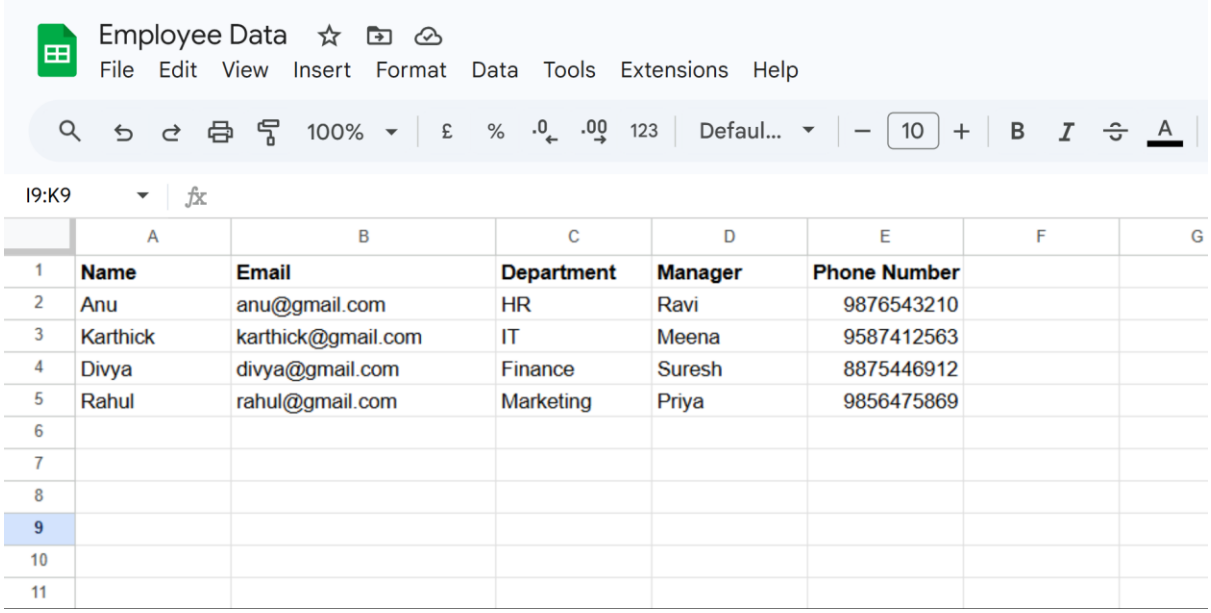
This step involves preparing structured employee data in a format that can be imported into ServiceNow.

Detailed Steps:

1. Open Google Sheets in your browser.
2. Create a new blank spreadsheet.
3. In the first row, define the column headers exactly as required:
    - Name
    - Email
    - Department
    - Manager
    - Phone Number
4. Enter sample employee records under each column.
    - Ensure email formats are correct
    - Maintain consistent naming for departments and managers
5. Rename the sheet as “Employee Data” for easy identification.
6. Click on File → Download and choose one of the supported formats:
    - Microsoft Excel (.xlsx) *(recommended)*
    - Comma Separated Values (.csv)

- XML

7. Save the file locally on your system.
    - This file will act as the data source for import



	A	B	C	D	E	F	G
1	**Name**	**Email**	**Department**	**Manager**	**Phone Number**		
2	Anu	anu@gmail.com	HR	Ravi	9876543210		
3	Karthick	karthick@gmail.com	IT	Meena	9587412563		
4	Divya	divya@gmail.com	Finance	Suresh	8875446912		
5	Rahul	rahul@gmail.com	Marketing	Priya	9856475869		
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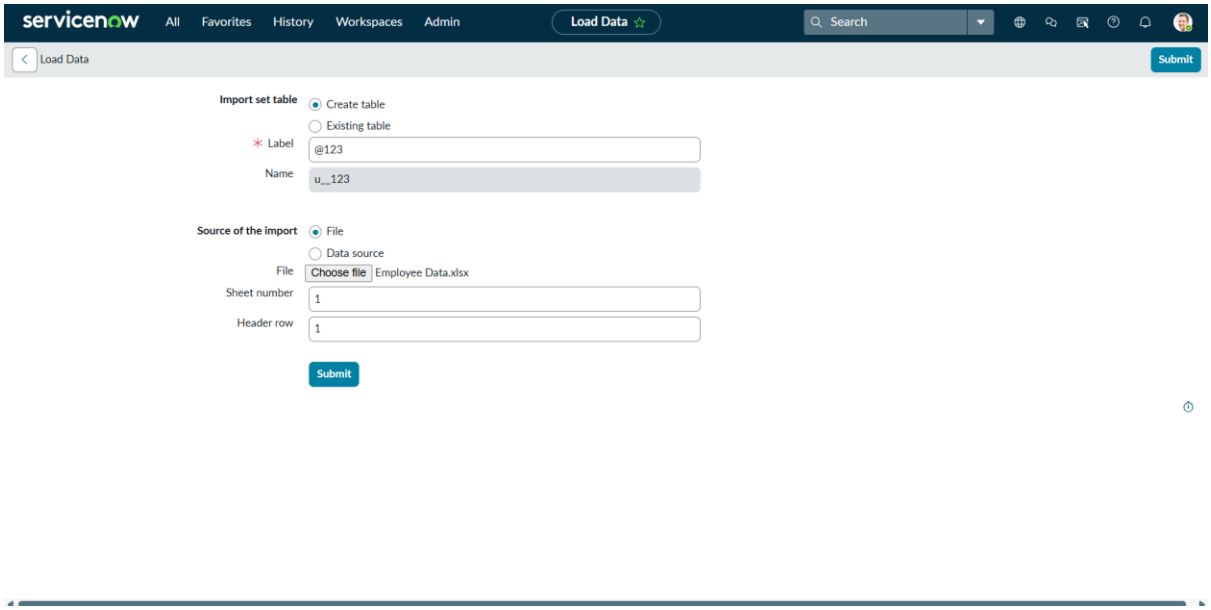
Milestone 2: Import Data Using Import Sets

Description:

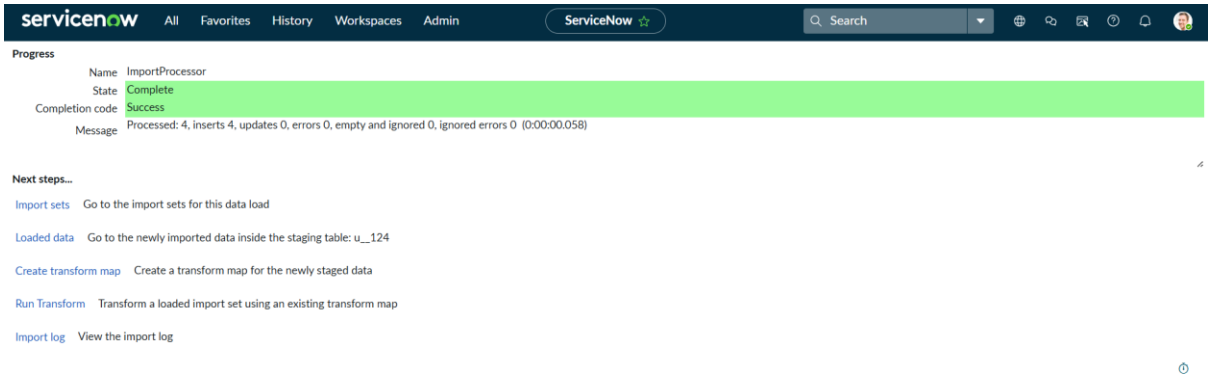
This stage involves importing the prepared data into ServiceNow using the Import Set framework, which ensures safe and controlled data migration.

Activity 1: Creating Import Table (Staging Table)

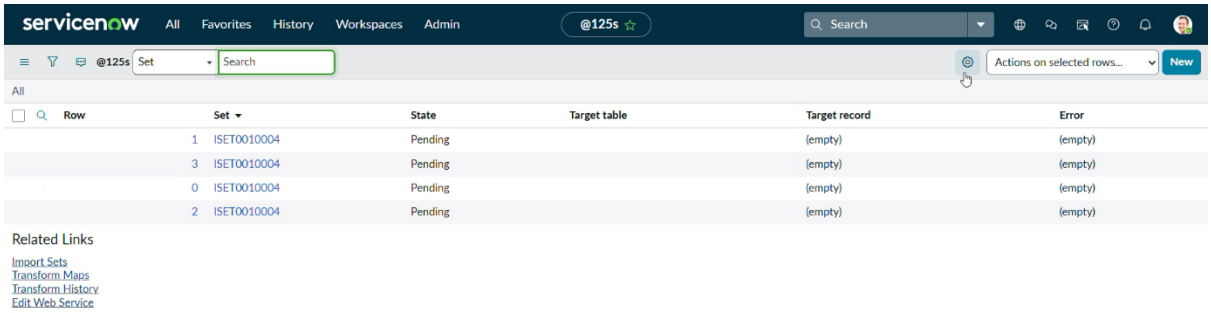
1. Log in to your ServiceNow instance.
2. In the left navigation panel, click on All.
3. Search for Import Sets.
4. Navigate to: System Import Sets → Load Data



5. Click on Create Table.
6. Enter the table label as: Import Employee
7. Click on Upload File and select the previously downloaded file.

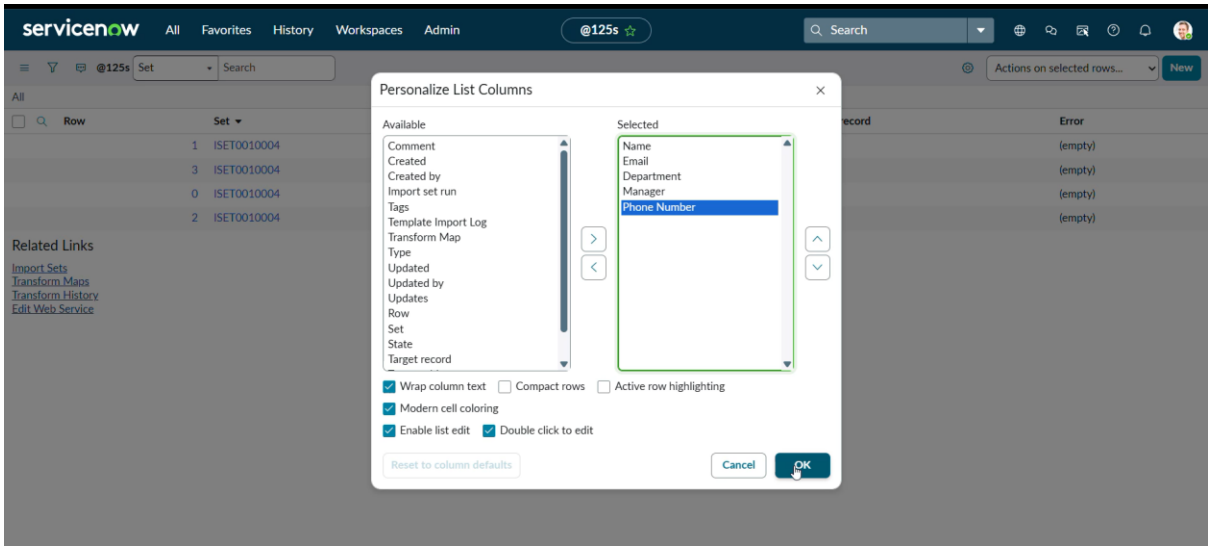


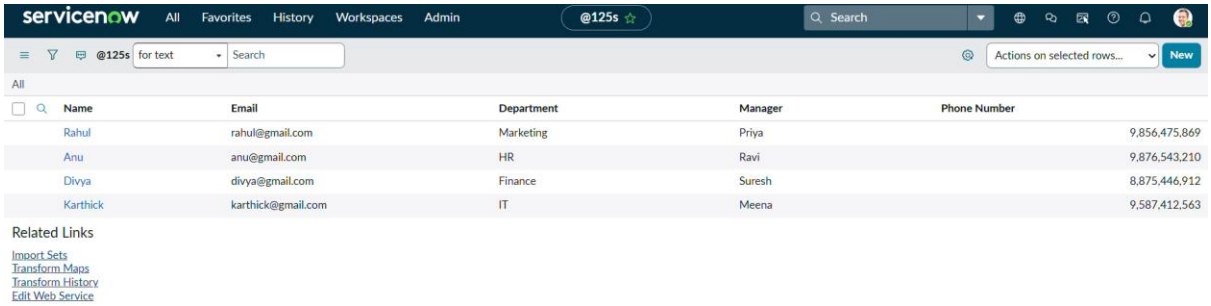
8. Click Submit to load the data.
9. Once the data is loaded:
    - A staging table (Import Set table) is automatically created
    - The imported records are displayed



10. Click on Personalize List (gear icon).

11. Add relevant fields to view all imported data clearly.

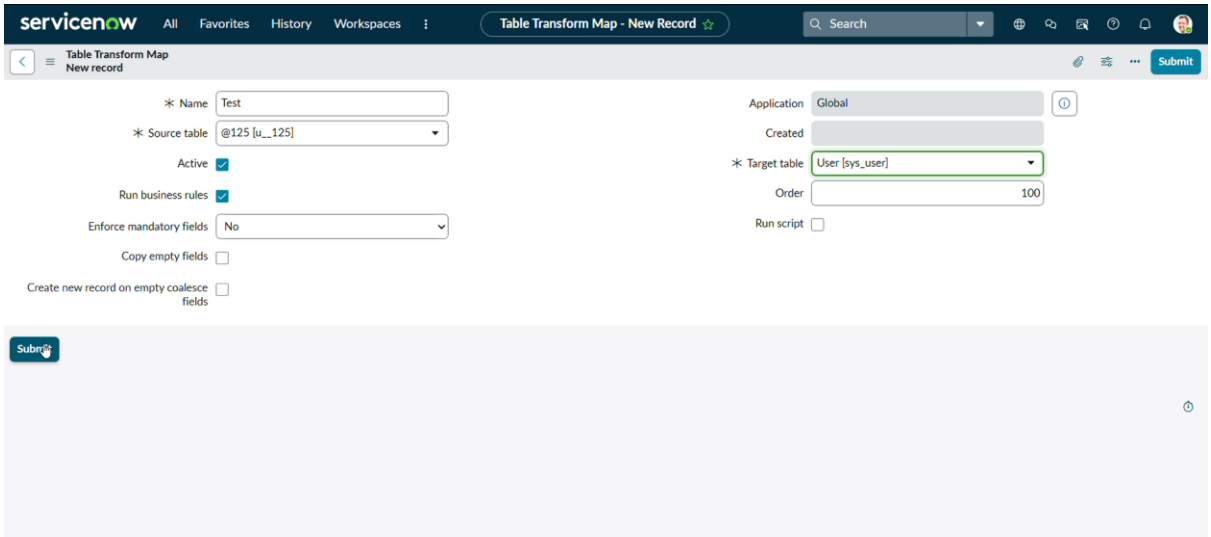




This staging table acts as a temporary storage area before moving data into actual system tables.

Activity 2: Creating Transform Map

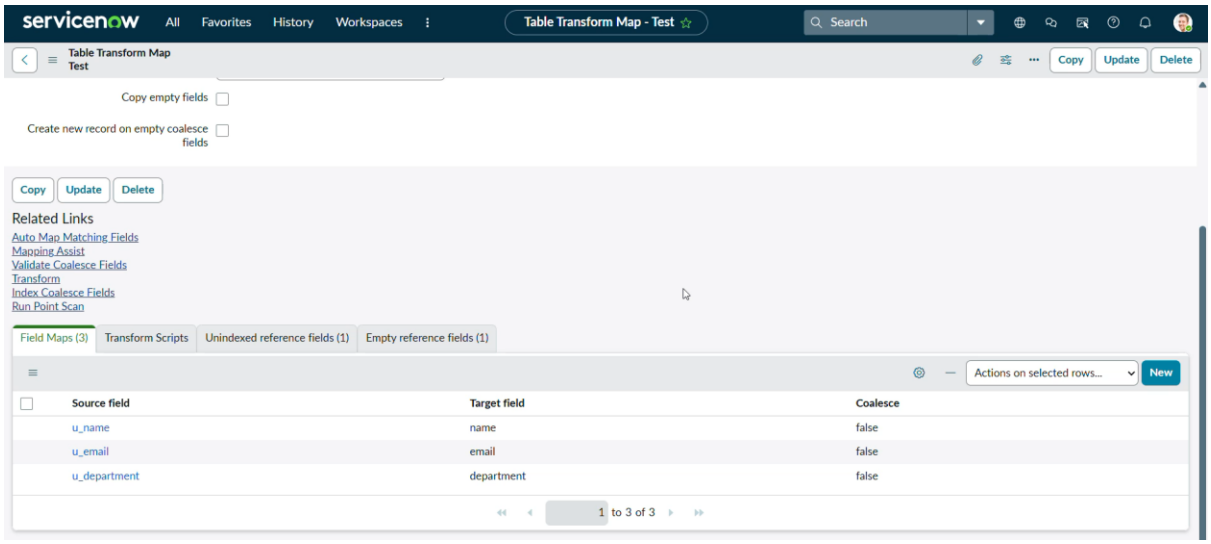
1. In the navigation bar, search for Transform Maps.
2. Navigate to: System Import Sets → Transform Maps
3. Click New to create a new Transform Map.
4. Fill in the details:
   - Name: Test
   - Source Table: Import Employee (your staging table)
   - Target Table: User [sys_user]
5. Click Save.



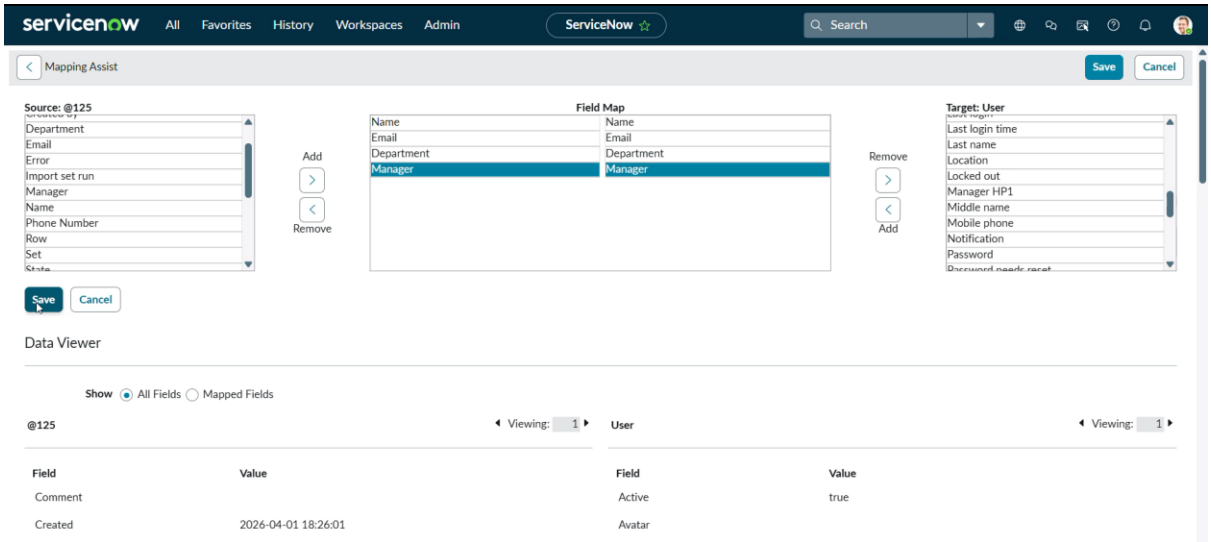
Transform Map defines how data moves from the staging table to the target table.

Activity 3: Match the Field Using Mapping Assist

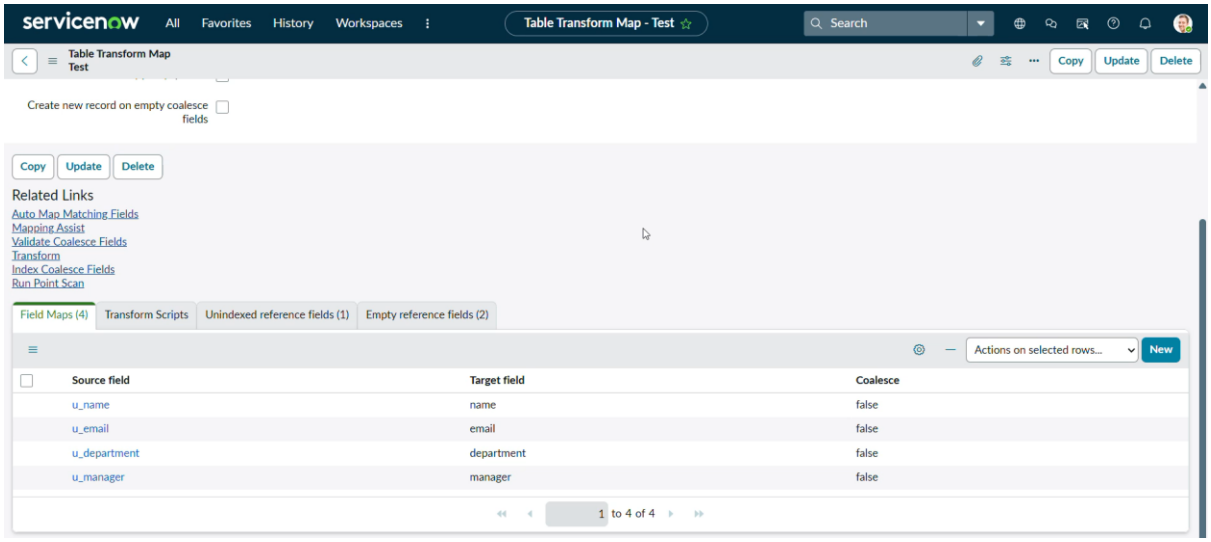
1. Open the created Transform Map.
2. Scroll down to the Field Maps related list.
3. Click on Auto Map Matching Fields:
    - This automatically maps fields with similar names



4. Review mapped fields carefully.
5. For fields that are not mapped (e.g., Manager):
    - Click Mapping Assist
    - Select:
        - ✓ Source Field: Manager
        - ✓ Target Field: Manager (reference field in sys_user)



6. Confirm the mapping and click Save.

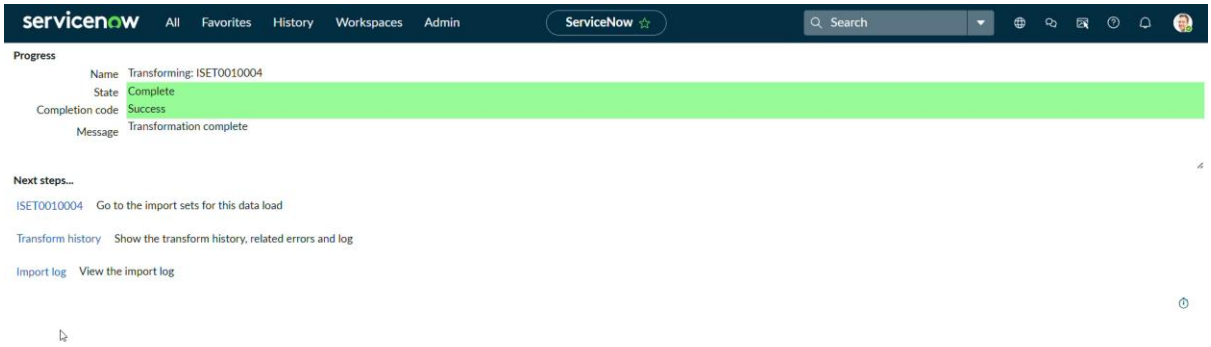


Proper mapping ensures correct data placement and maintains relationships.

Activity 4: Transform the Data

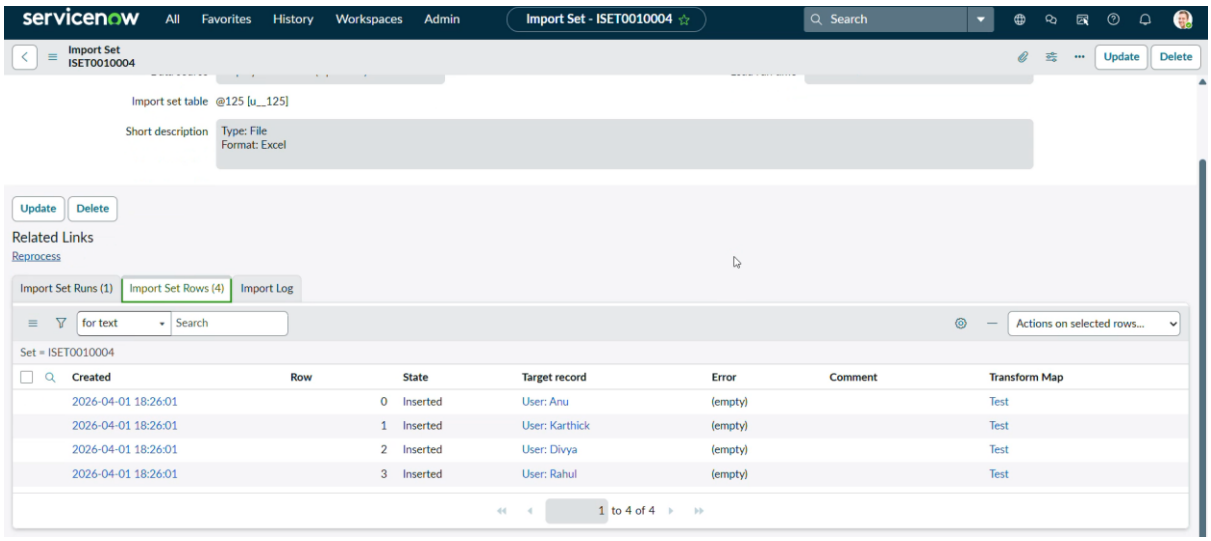
1. Open the Transform Map record.
2. In Related Links, click Transform.
3. Click Transform again to execute.
4. Monitor the process:

- Status should show Completed



5. Click on the Import Set Run record (e.g., ISETxxxx).
6. Verify:

- Number of records inserted
- No errors in transformation



At this stage, data is successfully moved to the User [sys_user] table.

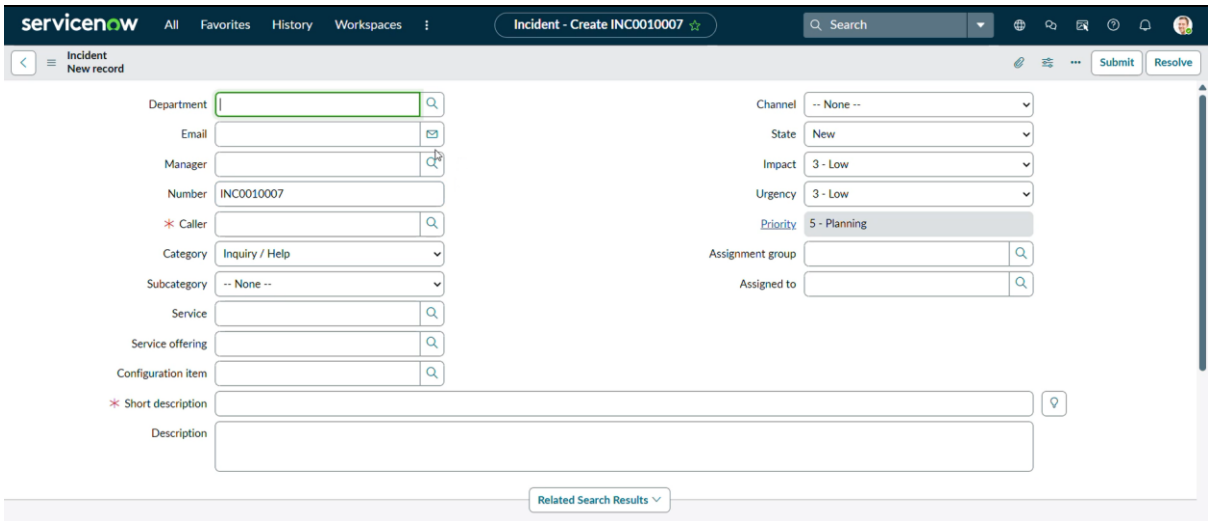
Milestone 3: Dot Walking

Description:

Dot Walking allows accessing related table fields through reference fields, improving usability without complex queries.

Activity 1: Display department to the assigned to user

1. In ServiceNow, navigate to: All → Incident → Create New
2. Observe the form:
    - Fields like Department, Email, Manager are not visible
3. Click on the Context Menu (☰ icon)
4. Select: Configure → Form Layout
5. In the available fields section:
    - Search and add:
        - Assigned To → Department
        - Assigned To → Email
        - Assigned To → Manager
6. Move these fields to the selected section.
7. Click Save.



Testing Dot Walking

8. Open a new Incident form.
9. Fill mandatory fields such as:
    - Short Description
    - Category
10. In Assigned To, select a user (imported earlier).
11. Observe:
    - Department auto-populates

- Email auto-populates
- Manager auto-populates

This confirms dot walking is working correctly.

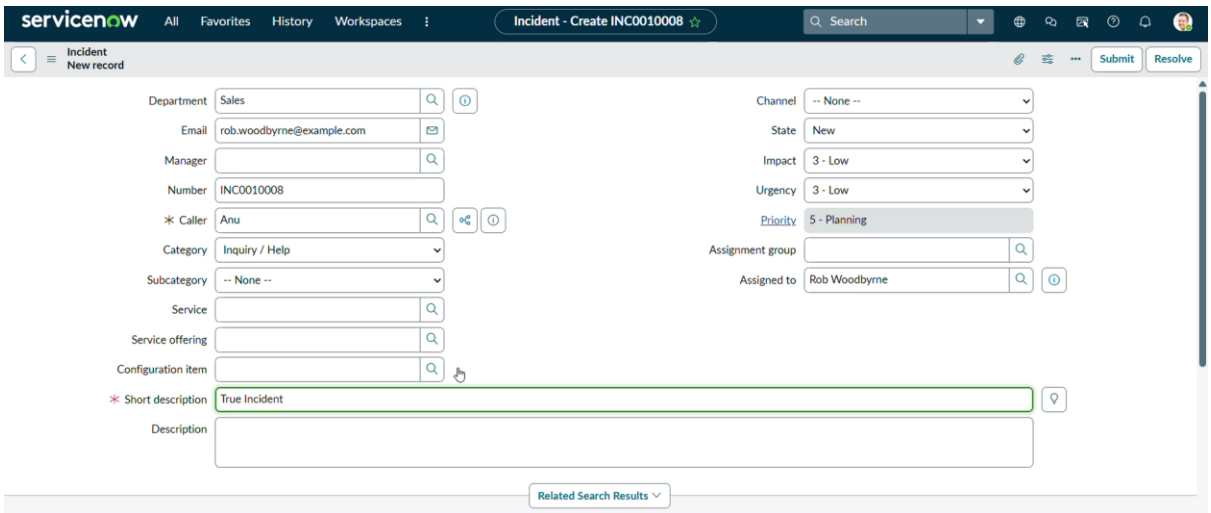
Milestone 4: Testing

Description:

Testing validates that the system behaves as expected after implementation.

Activity 1: Result

1. Navigate to Incident → Create New
2. Select a user in Assigned To field
3. Verify automatically populated fields:
    - Email
    - Department
    - Manager
4. Cross-check data accuracy with imported records



Successful testing confirms proper data import and relationship mapping.

Milestone 5: Conclusion

Description:

This project successfully automates data import and relationship mapping in ServiceNow.

Final Summary

- Import Sets provide a safe staging environment
- Transform Maps ensure accurate data mapping
- Reference fields maintain relationships
- Dot Walking simplifies data access

The implementation improves:

- Data accuracy
- System efficiency
- User experience

Conclusion

This project demonstrates a structured and efficient approach to automating data import and relationship mapping in ServiceNow. By leveraging Import Sets and Transform Maps, external data from sources such as Excel or CSV files can be seamlessly integrated into the system while maintaining data accuracy and consistency.

The use of Transform Maps ensures that data is correctly mapped from the staging table to the target table, while reference fields enable proper linkage between related records such as users, departments, and managers. Additionally, the implementation of dot walking enhances usability by allowing quick access to related information without the need for complex queries or manual navigation.

Overall, the solution reduces manual effort, minimizes errors, and improves operational efficiency. It provides a scalable and reliable method for handling large volumes of data while ensuring seamless relationship mapping across tables within the ServiceNow environment.